

Martingale 代表 未来条件期望 = 当下。大多不等式 为 已知波动且固定下，离原点之机率递减

但在 continuous monitoring, online learning, adaptive decision, 随机波动大小非恒定，且整体 Σ 也是未知且波动。

$\exists$ 1 个 gaussian Σ 有尾衰减不慢於它

Let filtration $(\Omega, \mathcal{F}, \mathcal{F}_{t \geq 0}, \mathbb{P})$, $\{Z_t\}_{t=1}^{\infty}$ be conditional sub gaussian martingale seq., $\{X_t\}_{t=1}^{\infty}$ is a predictable $\mathbb{R}^d$ random seq. define.

$$\sqrt{t} = \sum_{s=1}^t X_s \cdot Z_s \in \mathbb{R}^d$$

$$\sqrt{t} = \sqrt{I} + \sum_{s=1}^t X_s X_s^T \in \mathbb{R}^{d \times d} \quad (\sqrt{I} \text{ is an arbitrary positive definite matrix})$$

then $\forall \delta \in (0, 1)$, then take $1 - \delta$ probability $\Sigma \rightarrow \sqrt{t} \Sigma \sqrt{t}$ satisfies.

$$\mathbb{P} \left[\|\sqrt{t} \sqrt{t}\|_{\sqrt{t}} \leq \sqrt{2 \ln \left(\frac{\det(\sqrt{t} \sqrt{t})}{\det(\Sigma)} \right)} \right] = 1 - \delta.$$

$$\text{其中 } \Sigma > 0 \quad \Sigma \rightarrow \sqrt{\lambda} \in \mathbb{R} \quad \Sigma \rightarrow \mathbb{E}[\exp[\lambda \cdot Z_t] | \mathcal{F}_{t-1}] \leq \exp\left[\frac{\lambda^2 \Sigma}{2}\right]$$

這裡 Z_t 代表 martingale difference
表示隨機噪音 $\Sigma \rightarrow \mathbb{E}[Z_t | \mathcal{F}_{t-1}] = 0$.

$\sqrt{I}$ 表示先驗正則化矩陣. $\sqrt{I} \triangleq \lambda \cdot \text{Id} \quad (\lambda > 0)$

$\sqrt{t} = \sqrt{I} + \sum_{s=1}^t X_s X_s^T$ 為經驗斜方差

$$\|\sqrt{t}\|_{\sqrt{t}} = \sqrt{\sqrt{t} \cdot \sqrt{t} \cdot \sqrt{t}}$$

δ 全失敗機率.